# Low-Light Traffic Sign Detection Using GAN-Based Image Enhancement and YOLO

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***Abstract*—Low-light environments such as nighttime roads, tunnels, foggy scenes, and poorly illuminated traffic areas create major challenges for computer vision systems. In these conditions, images usually suffer from low contrast, strong noise, color distortion, and loss of structural details. These degradations reduce the performance of object detection models, especially YOLO-based detectors, because traffic signs and other road objects may lose clear edges, textures, and color information. This paper proposes a two-stage pipeline for low-light traffic sign detection using GAN-based image enhancement and YOLO-based object detection. In the first stage, a low-light image is passed through a Generative Adversarial Network (GAN) enhancement module to restore illumination, improve contrast, reduce noise, and preserve important object boundaries. In the second stage, the enhanced image is used as input for a YOLO detector to detect traffic signs and traffic-related objects. The system is evaluated using image enhancement metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), as well as detection metrics such as mean Average Precision (mAP), Precision, Recall, and FPS. By comparing YOLO performance on raw low-light images and GAN-enhanced images, this study aims to demonstrate the effectiveness of adding an enhancement module before object detection.**

***Index Terms*—Low-light image enhancement, GAN, YOLO, traffic sign detection, object detection, image preprocessing, computer vision.**

## I. Introduction

Low-light image conditions are common in real-world traffic environments. Nighttime roads, tunnels, rainy weather, foggy scenes, and poorly illuminated streets often produce images with weak illumination and poor visibility. In these situations, cameras may capture images with low contrast, strong sensor noise, color distortion, and missing structural details. These problems make it difficult for both humans and computer vision systems to understand the scene correctly.

Object detection models such as YOLO rely heavily on clear object boundaries, textures, shapes, and color information [7], [8]. However, in low-light images, important features of traffic signs may become unclear or disappear into the background. As a result, the detector may miss traffic signs, produce wrong bounding boxes, or generate low confidence scores. This issue is especially important in autonomous driving and intelligent transportation systems, where traffic sign detection must be accurate and reliable.

To address this problem, this paper proposes a two-stage pipeline. The first stage uses a GAN-based image enhancement network to improve the quality of low-light images. The second stage uses a YOLO-based object detector to detect traffic signs and other traffic-related objects from the enhanced images. The main idea is that better image visibility can improve the quality of visual features extracted by the detector.

GAN-based low-light image enhancement has been studied in previous works because adversarial learning can help generate visually realistic enhanced images [1], [2]. In this project, the enhancement module is used as a preprocessing step before YOLO detection.

The main contributions of this paper are summarized as follows:

- We propose a two-stage pipeline that combines low-light image enhancement and traffic object detection.
- We apply a GAN-based enhancement network with an encoder-decoder Generator, residual blocks, skip connections, and a PatchGAN Discriminator.
- We compare YOLO detection performance on raw low-light images and GAN-enhanced images.
- We evaluate the system using both image quality metrics and object detection metrics, including PSNR, SSIM, mAP, Precision, Recall, and FPS.

## II. Related Work

### A. Low-Light Image Enhancement

Low-light image enhancement aims to improve the visual quality of images captured under poor illumination. Traditional methods include histogram equalization, gamma correction, CLAHE, and Retinex-based algorithms. These methods can improve brightness and contrast, but they may also amplify noise, distort colors, or create unnatural lighting artifacts.

Deep learning methods have become more effective for low-light enhancement because they can learn complex mappings from dark images to normal-light images. These models can recover illumination, suppress noise, and preserve image structures more effectively than simple image processing techniques. Zero-DCE is one example of a deep learning-based method that enhances low-light images without requiring paired reference images [3].

### B. GAN-Based Image Enhancement

Generative Adversarial Networks (GANs) are widely used for image-to-image translation tasks. A GAN usually consists of two networks: a Generator and a Discriminator. The Generator learns to produce enhanced images from low-light inputs, while the Discriminator learns to distinguish between real normal-light images and generated enhanced images. Through adversarial training, the Generator gradually improves its ability to produce realistic and visually pleasing results.

For low-light image enhancement, GANs are useful because they can improve brightness while preserving realistic textures and natural colors. EnlightenGAN shows that adversarial learning can be applied to low-light enhancement without strictly requiring paired supervision [2]. The implementation used in this project follows a GAN-based image enhancement design inspired by an encoder-decoder image-to-image translation structure [1], [10].

However, GAN training can also be unstable, and the generated images may contain artifacts, color noise, or overexposed regions if the model is not trained carefully.

### C. YOLO-Based Object Detection

YOLO is a popular object detection framework because it is fast and suitable for real-time applications. YOLO predicts bounding boxes, class labels, and confidence scores directly from an input image [7]. Newer YOLO versions, such as YOLOv8, provide practical tools for training, validation, and inference in object detection tasks [8].

However, YOLO performance depends on the quality of the input image. In low-light conditions, object boundaries and textures become unclear, which can reduce detection accuracy. Therefore, improving the input image before detection may help YOLO extract stronger and more reliable features.

### D. Enhancement-Detection Pipeline

A two-stage enhancement-detection pipeline can be used to improve detection performance in difficult lighting conditions. The first stage enhances the low-light image, and the second stage performs object detection on the enhanced image. This structure allows the enhancement model to act as a preprocessing module before YOLO.

The key question in this project is whether the enhancement module can improve the downstream detection metrics such as mAP and Recall. Therefore, the system is evaluated by comparing YOLO performance on raw low-light images and GAN-enhanced images.

## III. Proposed Methodology

### A. Overall System Pipeline

The proposed method follows a two-stage pipeline:

$$\text{Low-light Image} \rightarrow \text{GAN Enhancement} \rightarrow \text{YOLO Detection} \rightarrow \text{Output}$$

The detailed process is shown below:

1) A low-light traffic image is used as the input.
2) The image is resized to $256 \times 256$ pixels.
3) Pixel values are normalized into the range $[-1, 1]$.
4) The normalized image is passed into the GAN Generator.
5) The Generator produces an enhanced image with improved illumination and contrast.
6) The enhanced image is converted back to the normal RGB image format.
7) Optional post-processing, such as median filtering, can be applied to reduce noise artifacts.
8) The enhanced image is passed into a YOLO detector.
9) YOLO outputs bounding boxes, class labels, and confidence scores.
10) The results are evaluated using image enhancement metrics and object detection metrics.

$$\boxed{\text{Low-light Image} \rightarrow \text{Preprocessing} \rightarrow \text{GAN Enhancement} \rightarrow \text{Enhanced Image} \rightarrow \text{YOLO Detection} \rightarrow \text{Detection Output}}$$

Fig. 1. Overall pipeline of the proposed low-light traffic sign detection system.

### B. Image Enhancement Network

The image enhancement module is based on a GAN architecture. The GAN contains two main parts: the Generator and the Discriminator. The Generator converts low-light images into enhanced images, while the Discriminator evaluates whether the generated images look realistic.

*1) Generator Architecture:* The Generator follows a U-Net-like encoder-decoder architecture. The encoder extracts visual features from the low-light image, while the decoder reconstructs the enhanced image. Residual blocks are placed between the encoder and decoder to preserve important information and improve training stability. Skip connections are also used to transfer low-level spatial details from encoder layers to decoder layers.

The Generator pipeline can be summarized as:

$$\text{Input} \rightarrow \text{Encoder} \rightarrow \text{Residual Blocks} \rightarrow \text{Decoder} \rightarrow \text{Output}$$

The detailed architecture is shown below:

TABLE I
GENERATOR ARCHITECTURE.

| Stage | Description |
|---|---|
| Input | RGB low-light image with size $256 \times 256 \times 3$ |
| Encoder 1 | Conv2D with 64 filters, $7 \times 7$ kernel, BatchNorm, LeakyReLU |
| Encoder 2 | Conv2D with 128 filters, $3 \times 3$ kernel, stride 2, BatchNorm, LeakyReLU |
| Encoder 3 | Conv2D with 256 filters, $3 \times 3$ kernel, stride 2, BatchNorm, LeakyReLU |
| Bottleneck | Six residual blocks with 256 feature channels |
| Decoder 1 | UpSampling2D, Conv2D with 128 filters, Dropout, skip connection |
| Decoder 2 | UpSampling2D, Conv2D with 64 filters, Dropout, skip connection |
| Output | Conv2D with 3 filters, $7\times7$ kernel, Tanh activation |

The encoder is responsible for extracting multi-scale features such as edges, object contours, and illumination patterns. The residual blocks help reduce information loss during deep feature transformation. The decoder reconstructs the enhanced image by gradually increasing the spatial resolution. Skip connections help preserve high-frequency details, which are important for traffic sign detection.

*2) Discriminator Architecture:* The Discriminator uses a PatchGAN architecture. Instead of classifying the entire image as real or fake, PatchGAN evaluates small local patches of the image. This allows the Discriminator to focus on local texture quality, sharpness, noise, and small object details [10].

The Discriminator receives two images as input: the low-light source image and either the real target image or the generated enhanced image. These two images are concatenated into a $256 \times 256 \times 6$ tensor. The network then applies a sequence of convolutional layers and produces a patch-level classification map.

TABLE II
PATCHGAN DISCRIMINATOR ARCHITECTURE.

| Stage | Description |
|---|---|
| Input | Concatenated source and target/generated image, $256 \times 256 \times 6$ |
| Layer 1 | Conv2D with 64 filters, $4 \times 4$ kernel, stride 2, LeakyReLU |
| Layer 2 | Conv2D with 128 filters, $4 \times 4$ kernel, stride 2, BatchNorm, LeakyReLU |
| Layer 3 | Conv2D with 256 filters, $4 \times 4$ kernel, stride 2, BatchNorm, LeakyReLU |
| Layer 4 | Conv2D with 512 filters, $4 \times 4$ kernel, stride 2, BatchNorm, LeakyReLU |
| Layer 5 | Conv2D with 512 filters, $4\times4$ kernel, BatchNorm, LeakyReLU |
| Output | Conv2D with 1 filter, $4 \times 4$ kernel, Sigmoid activation |

Patch-level discrimination is useful for this project because traffic signs are often small objects. Local details such as edges, symbols, and sign boundaries can strongly affect YOLO detection performance.

### C. Loss Function

The current implementation uses adversarial loss and L1 reconstruction loss. The adversarial loss encourages the Generator to produce images that look realistic, while the L1 loss helps preserve the structure and content of the target image.

The total loss is defined as:

$$L_{total} = L_{adv} + \lambda L_1 \tag{1}$$

where $L_{adv}$ is the adversarial loss, $L_1$ is the L1 reconstruction loss, and $\lambda$ controls the importance of the reconstruction loss. In this implementation, the L1 loss weight is set to 100 [1].

The adversarial loss helps improve realism, while the L1 loss prevents the Generator from changing the image content too much. This is important because the enhanced image must still preserve traffic signs, road objects, and scene structures for object detection.

### D. YOLO-Based Traffic Sign Detection

After the enhancement stage, the enhanced images are passed into a YOLO-based object detector. YOLO outputs bounding boxes, class labels, and confidence scores for traffic objects. The main experiment is designed to compare two cases:

- YOLO detection on raw low-light images.
- YOLO detection on GAN-enhanced images.

The improvement in detection performance is measured using mAP, Precision, and Recall. If the GAN-enhanced images produce higher mAP and Recall than the raw low-light images, this indicates that the enhancement module improves the quality of the input features used by YOLO.

## IV. DATASETS

### A. LOL Dataset

The LOL Dataset is used for low-light image enhancement. It contains paired low-light and normal-light images. In this project, 485 image pairs are used for training and 15 image pairs are used for testing. Since the dataset provides paired low-light and normal-light images, it is suitable for evaluating image quality using PSNR and SSIM [4].

However, LOL does not contain bounding box annotations. Therefore, it is used only for the image enhancement stage and not for object detection evaluation.

### B. BDD100K Nighttime Subset

BDD100K is a large-scale driving dataset that contains images from real traffic scenes. In this project, the nighttime subset is used for traffic object detection evaluation. The extracted nighttime subset contains approximately 31,899 images. A recommended split is 26,422 images for training and 5,477 images for validation.

Due to time limitations, this project may not directly train YOLO on the full BDD100K nighttime training set. Instead, the validation or test subset can be used to compare YOLO performance before and after GAN enhancement.

BDD100K contains bounding box annotations for multiple traffic-related classes, such as car, bus, traffic sign, traffic light, person, rider, bike, motor, truck, and train. This makes it suitable for evaluating whether image enhancement improves downstream detection performance [5].

### C. ExDark Dataset

The ExDark Dataset contains 7,363 real low-light images captured under difficult lighting conditions. It includes 12 object classes and bounding box annotations. Compared with LOL, ExDark is more challenging because it contains severe darkness, low contrast, and strong real-world noise [6].

In this project, ExDark is used as a robustness test dataset. It helps evaluate whether the proposed enhancement-detection pipeline can work in more extreme low-light environments.

### D. LSRW Dataset

The LSRW Dataset contains real-world low-light and normal-light image pairs. It is useful for improving GAN training because it provides more diverse real-world low-light examples [9]. However, due to time limitations, LSRW is not used in the current stage of this project. It is planned as future work for the final version.

TABLE III
DATASET SUMMARY.

| Dataset | Size | Annotation | Used for |
|---|---|---|---|
| LOL | 500 paired images | Paired low-light and normal-light images, no bounding boxes | GAN training and PSNR/SSIM evaluation |
| BDD100K Nighttime | Around 31,899 images | Traffic object bounding boxes | YOLO comparison before and after enhancement |
| ExDark | 7,363 images | 12 object classes with bounding boxes | Robustness test under severe low-light conditions |
| LSRW | About 5K+ image pairs | Paired real-world low-light and normal-light images | Future GAN training |

## V. Experimental Setup

### A. GAN Training Configuration

The GAN enhancement model is implemented using TensorFlow and Keras. The input image size is set to $256 \times 256 \times 3$. The optimizer is Adam with a learning rate of 0.0002 and $\beta_1 = 0.5$. The model is trained using adversarial loss and L1 reconstruction loss [1].

TABLE IV
GAN TRAINING CONFIGURATION.

| Parameter | Value |
|---|---|
| Framework | TensorFlow / Keras |
| Input size | $256 \times 256 \times 3$ |
| Image normalization | $[-1, 1]$ |
| Optimizer | Adam |
| Learning rate | 0.0002 |
| Beta1 | 0.5 |
| Batch size | 12 |
| Epochs | 100 |
| Loss function | Adversarial loss + L1/MAE loss |
| L1 weight | 100 |
| Output activation | Tanh |

### B. Detection Configuration

The enhanced images are evaluated using a YOLO-based detector. The detector is tested on both raw low-light images and GAN-enhanced images. The goal is to measure whether the enhancement stage improves detection accuracy.

TABLE V
YOLO DETECTION CONFIGURATION.

| Parameter | Value |
|---|---|
| Detector | YOLOv8 / YOLOv11 |
| Input type 1 | Raw low-light images |
| Input type 2 | GAN-enhanced images |
| Main dataset | BDD100K Nighttime subset |
| Robustness dataset | ExDark |
| Main classes | Traffic sign and traffic-related objects |
| Metrics | mAP, Precision, Recall, FPS |

## VI. Evaluation Metrics

### A. Image Enhancement Metrics

The quality of the enhanced images is evaluated using PSNR and SSIM.

*1) PSNR:* Peak Signal-to-Noise Ratio (PSNR) measures the pixel-level difference between the enhanced image and the ground-truth normal-light image. A higher PSNR value usually indicates better image reconstruction quality.

$$PSNR = 10 \log_{10} \left( \frac{MAX_I^2}{MSE} \right) \tag{2}$$

where $MAX_I$ is the maximum possible pixel value and $MSE$ is the mean squared error between the enhanced image and the reference image.

*2) SSIM:* Structural Similarity Index Measure (SSIM) measures the structural similarity between two images. Unlike PSNR, SSIM considers luminance, contrast, and structural information. A higher SSIM value means the enhanced image preserves more structural details from the reference image.

### B. Object Detection Metrics

The detection performance is evaluated using mAP, Precision, Recall, and FPS.

*1) Precision:* Precision measures how many predicted objects are correct.

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

where $TP$ is true positive and $FP$ is false positive.

*2) Recall:* Recall measures how many real objects are successfully detected.

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

where $FN$ is false negative.

*3) mAP:* Mean Average Precision (mAP) is the main metric for object detection. It summarizes detection performance across object classes and confidence thresholds. In this project, mAP is used to compare YOLO performance on raw low-light images and GAN-enhanced images.

*4) FPS:* Frames per second (FPS) measures inference speed. This metric is important because the proposed system adds an enhancement model before YOLO. If the pipeline is intended for real-time traffic applications, FPS must be considered.

## VII. Results and Benchmark

### A. Image Enhancement Benchmark

The image enhancement benchmark compares the quality of raw low-light images and GAN-enhanced images. PSNR and SSIM are calculated on paired datasets such as LOL.

TABLE VI
Image enhancement benchmark.

| Method | Dataset | PSNR ↑ | SSIM ↑ |
|---|---|---|---|
| Raw low-light image | LOL Test | – | – |
| GAN Enhancement | LOL Test | To be filled | To be filled |
| GAN + Post-processing | LOL Test | To be filled | To be filled |

### B. Detection Benchmark

The detection benchmark compares YOLO performance on raw low-light images and GAN-enhanced images. This benchmark is the most important part of the project because it shows whether the enhancement module improves detection accuracy.

### C. Qualitative Results

In addition to quantitative metrics, visual comparison is also important. The qualitative results should show examples of low-light images, GAN-enhanced images, YOLO detection on raw images, and YOLO detection on enhanced images.

Insert visual comparison here:
Raw low-light image — GAN-enhanced image
— YOLO on raw — YOLO on enhanced

Fig. 2. Qualitative comparison between raw low-light images and GAN-enhanced detection results.

The expected observation is that GAN-enhanced images provide better visibility, clearer object boundaries, and stronger contrast. These improvements may help YOLO detect traffic signs with higher confidence. However, if the enhancement model creates artifacts, overexposure, or color noise, the detection result may not always improve.

## VIII. Implementation Challenges

Several difficulties occurred during the training and evaluation process.

### A. GAN Training Instability

GAN training is more difficult than normal supervised learning because the Generator and Discriminator must be trained together. If the Discriminator becomes too strong, the Generator may not receive useful gradients. If the Generator becomes too strong, the Discriminator may fail to provide meaningful feedback. This can cause unstable losses and inconsistent image quality.

### B. Color Noise and Over-Enhancement

Some enhanced images may become brighter but also contain color noise or unnatural color shifts. In some cases, bright regions may become overexposed. This is a problem because YOLO may rely on color and edge information to detect traffic signs. If enhancement changes the object appearance too much, detection performance may decrease.

### C. Dataset Pairing Problem

The LOL Dataset provides paired low-light and normal-light images, so it is useful for PSNR and SSIM evaluation. However, datasets such as BDD100K and ExDark contain detection labels but do not always provide paired normal-light reference images. Therefore, PSNR and SSIM cannot be directly calculated on all detection datasets. For this reason, detection metrics such as mAP and Recall are more important for evaluating the full pipeline.

### D. Domain Shift

There is a domain difference between enhancement datasets and detection datasets. LOL contains paired low-light images, but it may not fully represent nighttime traffic scenes. BDD100K and ExDark contain more realistic driving or outdoor low-light scenes. Because of this domain shift, a GAN trained on LOL may not always generalize well to traffic sign detection images.

### E. Annotation Conversion

BDD100K annotations need to be converted into YOLO format before detection evaluation or training. This process can be difficult because image names, class IDs, and bounding box coordinates must match correctly. Any error in annotation conversion can cause incorrect evaluation results.

TABLE VII
YOLO DETECTION PERFORMANCE BEFORE AND AFTER IMAGE ENHANCEMENT.

| Input Type | Dataset | mAP@0.5 ↑ | Precision ↑ | Recall ↑ | FPS ↑ |
|---|---|---|---|---|---|
| Raw low-light | BDD100K Nighttime | To be filled | To be filled | To be filled | To be filled |
| GAN-enhanced | BDD100K Nighttime | To be filled | To be filled | To be filled | To be filled |
| Raw low-light | ExDark | To be filled | To be filled | To be filled | To be filled |
| GAN-enhanced | ExDark | To be filled | To be filled | To be filled | To be filled |

### *F. Image Naming and Path Bugs*

Another practical difficulty is dataset organization. The image file names, ground-truth image names, and label file names must match correctly. If file names are inconsistent, the training or evaluation script may fail to load the correct image pairs. This issue is especially important when using multiple datasets.

### *G. Inference Speed*

Adding a GAN enhancement model before YOLO increases the total processing time. This can reduce FPS and make real-time deployment more difficult. Therefore, inference speed must be measured in addition to accuracy.

## IX. DISCUSSION

The main goal of this project is not only to generate visually brighter images, but also to improve downstream traffic sign detection. If YOLO achieves higher mAP and Recall on GAN-enhanced images than on raw low-light images, this means the enhancement module helps the detector extract better features.

However, image enhancement does not always guarantee better detection. An enhanced image may have higher brightness but still contain artifacts, color distortion, or blurred details. These issues can confuse the detector. Therefore, the evaluation must include both image quality metrics and detection metrics.

PSNR and SSIM are useful for measuring image restoration quality on paired datasets such as LOL. However, mAP, Precision, and Recall are more important for measuring the real value of the system because the final task is object detection.

## X. LIMITATIONS

The current project still has several limitations. First, the GAN model is mainly trained on paired enhancement data and may not fully generalize to all nighttime traffic scenes. Second, YOLO has not yet been fully fine-tuned on GAN-enhanced images. Third, the LSRW dataset has not been used in the current training stage due to time limitations. Fourth, the pipeline may be slower than using YOLO alone because it adds an enhancement stage before detection. Finally, the GAN may create artifacts that reduce detection performance in some cases.

## XI. FUTURE WORK

Several improvements are planned for the final version of the project. First, the GAN model will be trained on the LSRW dataset to improve real-world generalization. Second, hyperparameters will be fine-tuned to reduce color noise and overexposure. Third, YOLO will be trained or fine-tuned directly on GAN-enhanced images so that the detector can adapt to the enhanced image domain. Fourth, the system will be optimized to improve FPS for possible real-time usage. Finally, an ablation study can be added to analyze the contribution of each component.

TABLE VIII
FUTURE IMPROVEMENT PLAN.

| Improvement | Purpose | Expected Impact |
|---|---|---|
| Train GAN on LSRW | Improve real-world low-light enhancement | Better generalization and image quality |
| Fine-tune hyperparameters | Reduce color noise and overexposure | Higher PSNR and SSIM |
| Fine-tune YOLO on enhanced images | Adapt detector to enhanced domain | Higher mAP and Recall |
| Optimize FPS | Improve inference speed | Better real-time performance |
| Add ablation study | Evaluate each module separately | Stronger experimental analysis |

## XII. CONCLUSION

This paper presents a two-stage pipeline for traffic sign detection under low-light conditions. The first stage uses a GAN-based enhancement network to improve illumination, contrast, and structural details in low-light images. The second stage uses a YOLO-based detector to identify traffic signs and traffic-related objects. The system is evaluated using both image enhancement metrics and object detection metrics. The main objective is to determine whether GAN-enhanced images can improve YOLO detection performance compared with raw low-light images. Although the current system still has limitations such as domain shift, possible color artifacts, and additional inference time, the proposed pipeline provides a practical direction for improving object detection in nighttime and low-light traffic environments.

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